



Software Requirements Specifications for Sif (Autonomous farming system)

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Prepared By
Specialized
Automated
Solutions

Change History

Date	Version	Description	Updated By
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Document Approvals

Name	Role	Signature	Version
Bryan Martinez	Team leader		0.1
Bryan Martinez			1.0

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1 Introduction

1.1 Purpose

The purpose of this document is to describe the business requirements in details which are necessary for the implementation of the application. All the business requirement must be reflected in a version of this document in order to be included in the final implementation. This document also details out the nonfunctional and technical requirements. Also necessary technical designs which will include each of the business requirements.

1.2 Document Conventions

This document uses the following conventions.

DB	Database
DDB	Distributed Database
ER	Entity Relationship
GCM	General Control Module
RCM	Rover Control Module
GCM	Greenhouse Control Module

1.3 Project Scope

This project is a complete integer system that allow the user to manage remotely the environment variables of a cultivation site also to set an autonomous behavior for the upcoming variables. Added to this the system will automatically store all the data so the user can find or decide a phat to optimize the cultivation of the edible.

1.4 Intended Audience

This project is an optimization for veteran farmers or users that want to investigate the optimal variables to cultivate an edible or simple to automate the farming process of some of them. This project is useful for the agronomist community.

1.5 References

Full documents SOW, SOC prototype, etc. are annexed

2 System Description

2.1 Current System:

Currently the user only has a rudimentary irrigation system in a greenhouse, it only has the capability of automate the irrigation to some intervals of time.

2.2 Proposed System:

There should be an app that allow the user to set the environment variables and wished behavior of the systems. After the user set the behavior of the systems it will star to operate autonomously. At the same time the app will allow the user to access the stored data an to give some instructions to the system to adjust variables or execute manual modules.

The system can be a statistical tool the user can use to measure the grown of the edible compared with the environment variables set and then replicate to optimize thr production.

2.3 Proposed System High level design:

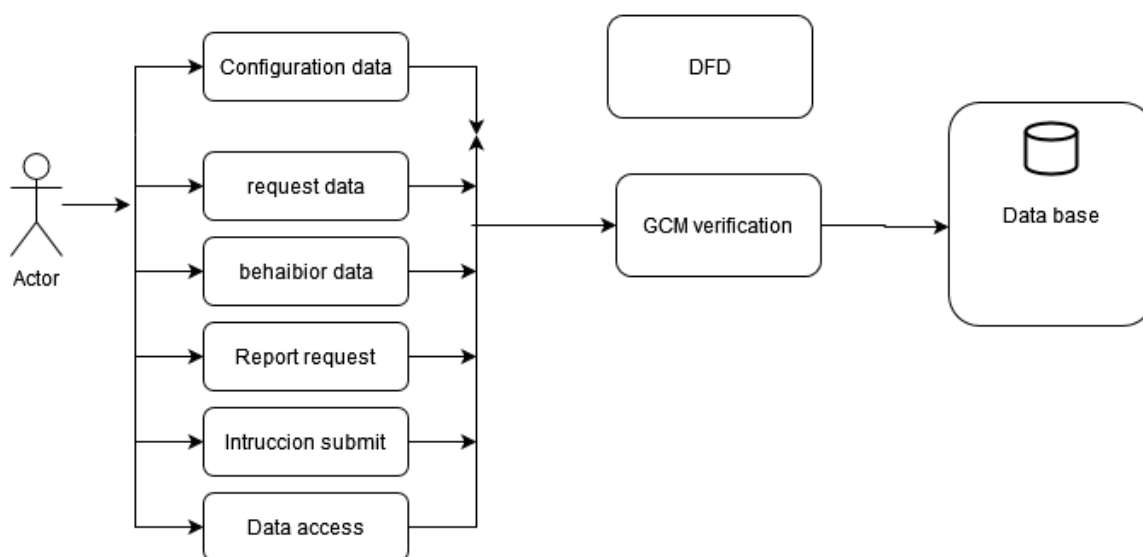
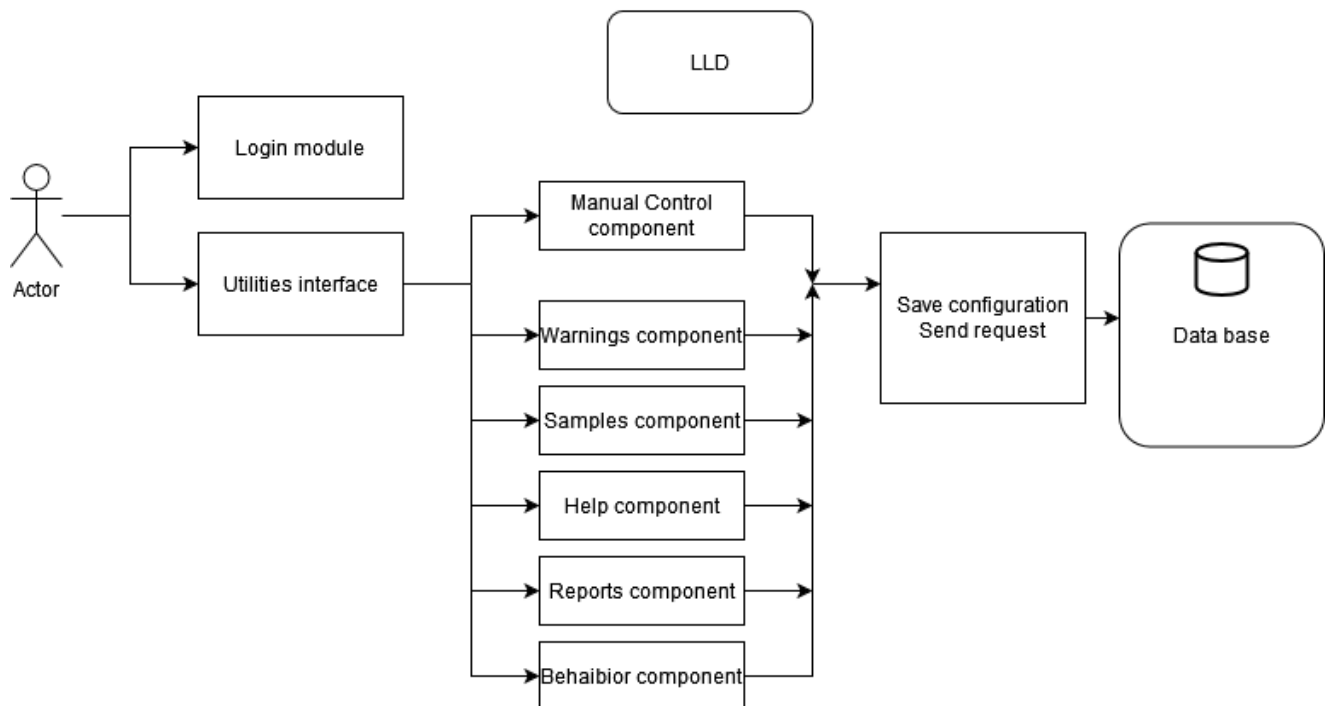
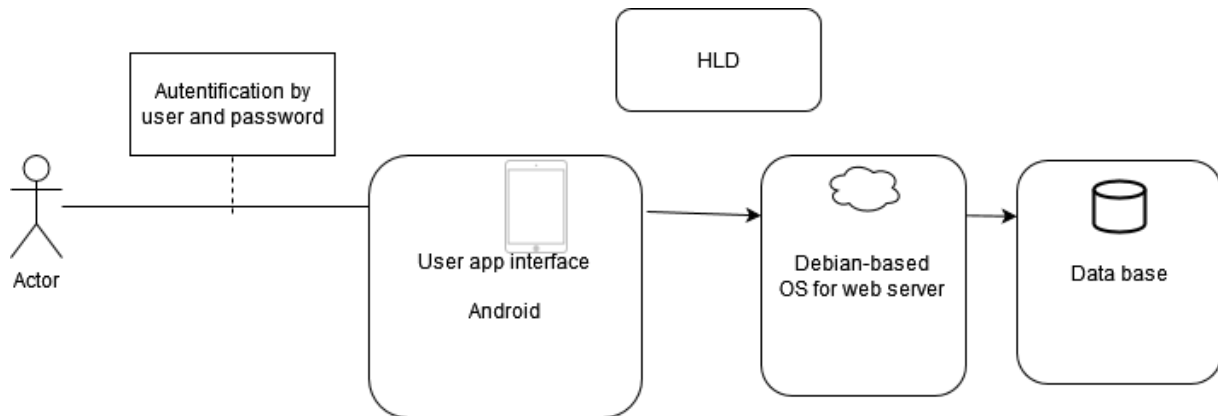
High level design will be deploy once project has a patron or already 40% of investment.

3 Functional Requirements

3.1 System Features

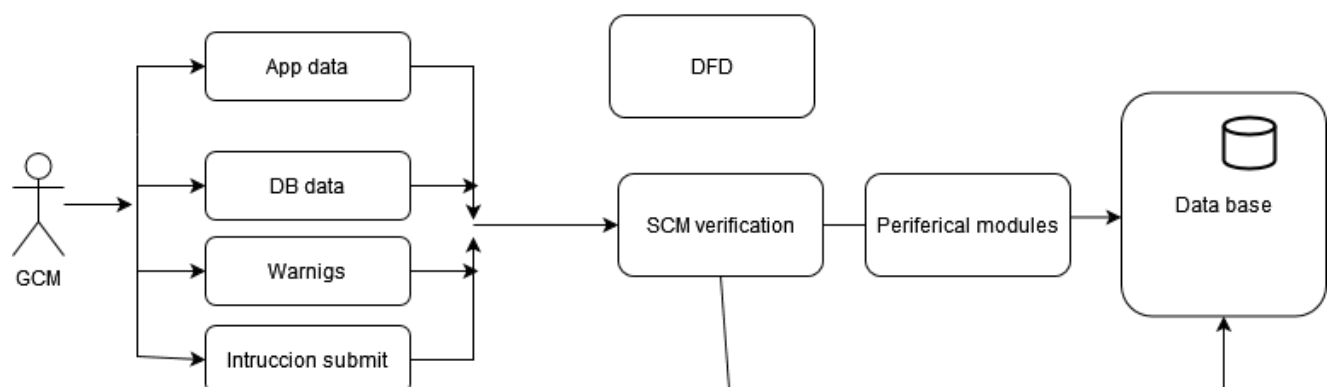
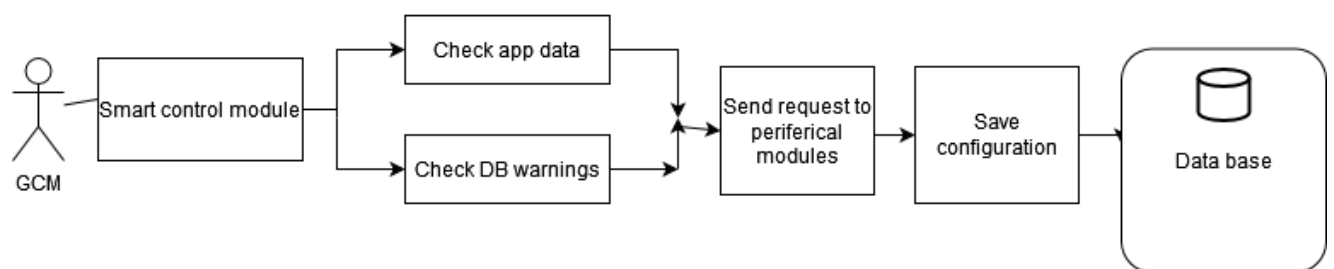
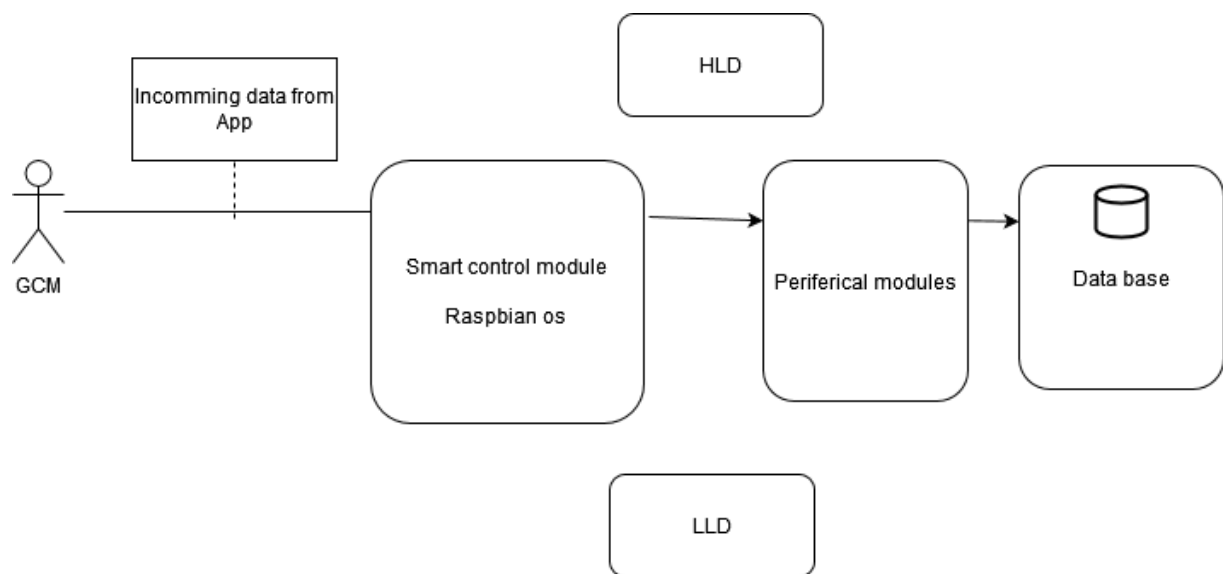
3.1.1 User app

User app	
App will receive data and instructions from user also will give access to stored data in server.	
Inputs	Configuration data instructions for modules Behavior changes access to system
outputs	Show stored data Show warnings



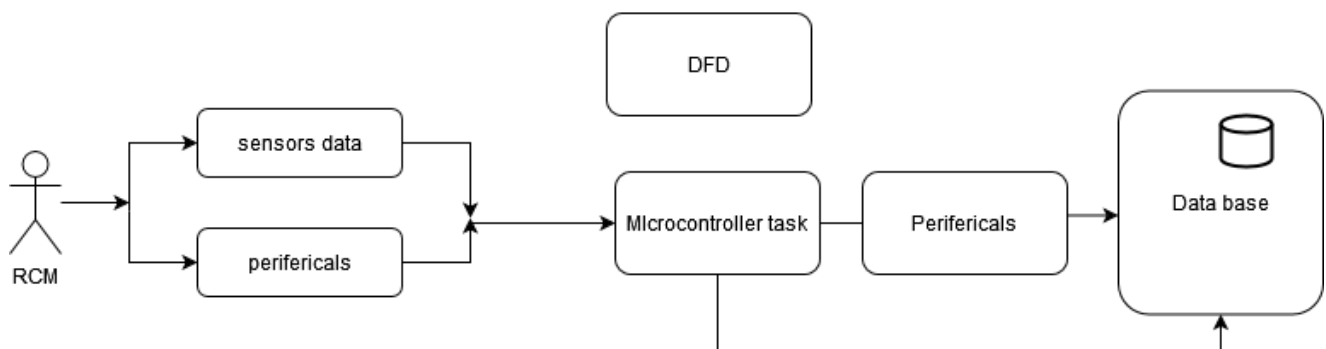
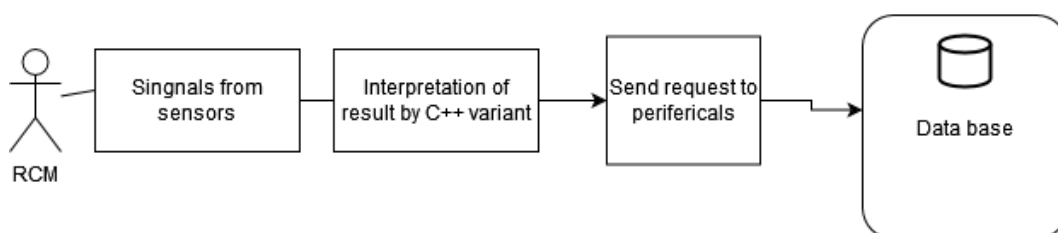
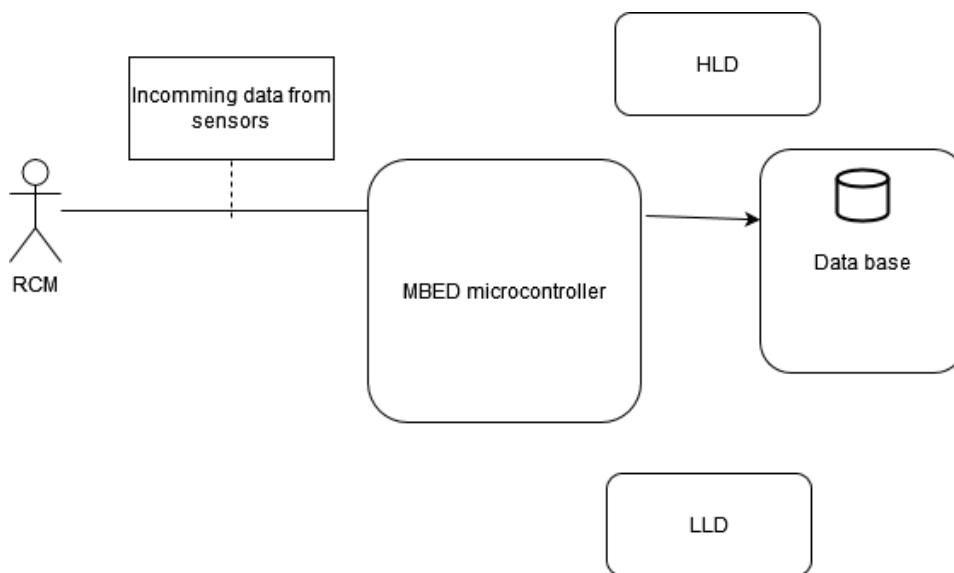
3.1.2 General Control Module (GCM).

GCM	
GCM will receive data from app and DB then transmit it to the needed module, also it would change the behavior of some modules depending of the data received from the DB or the app.	
<i>inputs</i>	<i>Data stored in the DB Instruction's from the user app</i>
<i>outputs</i>	<i>Instructions for external modules Behaviors for modules</i>



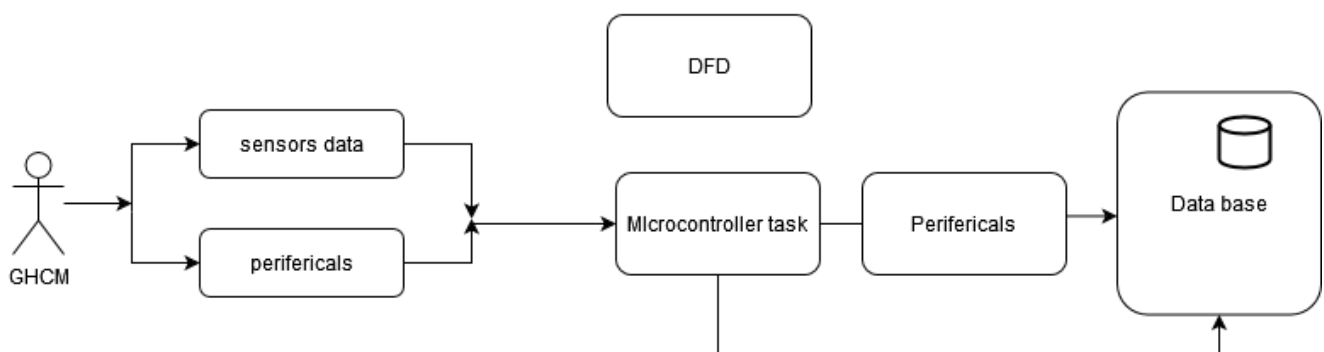
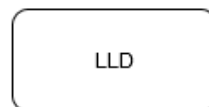
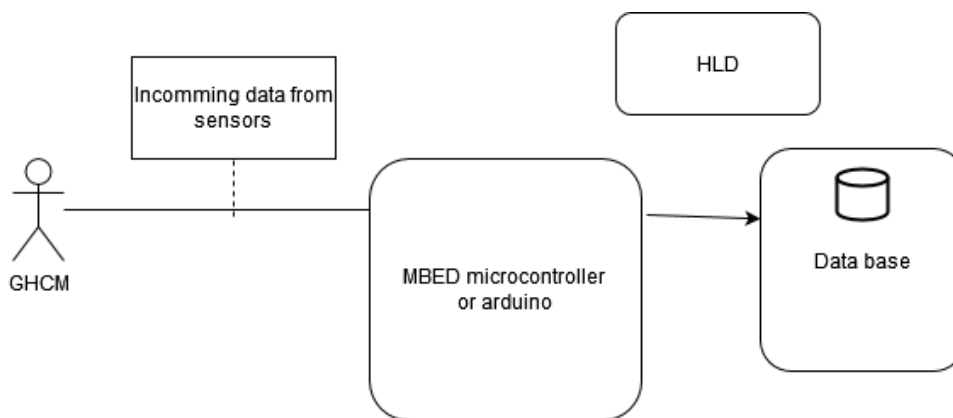
3.1.3 Rover Control Module (RCM).

RCM	
RCM will receive instructions form GCM and execute them, also receive data from sensors and send it to the DB.	
<i>inputs</i>	<i>Signals from sensors instructions from GCM</i>
<i>outputs</i>	<i>Processed data to storage</i>



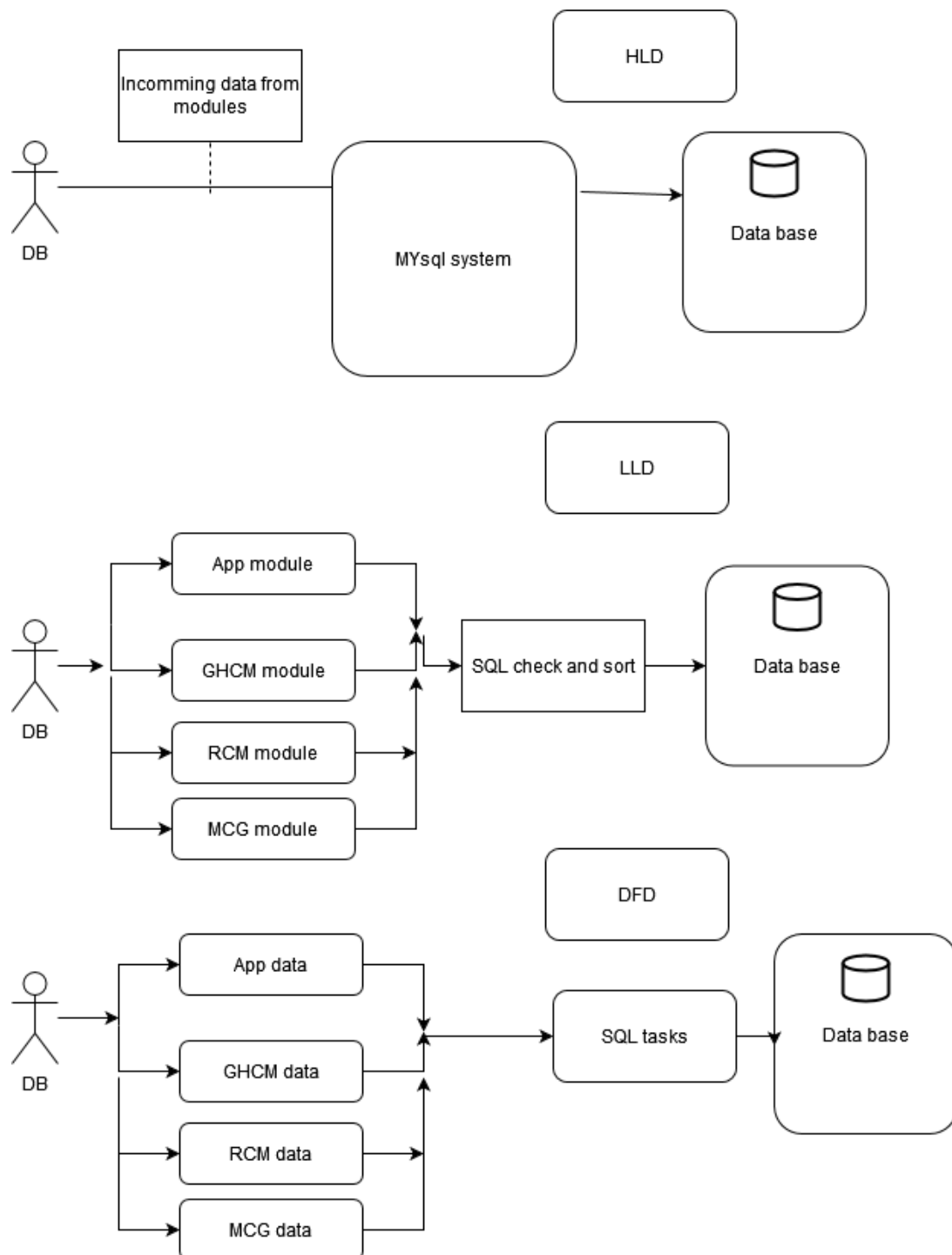
3.1.4 Greenhouse Control Module (GHCM).

GHCM	
GCM will receive instructions from GHCM and execute them, also receive data from sensors and send it to the DB.	
<i>inputs</i>	<i>Signals from sensors instructions from GCM</i>
<i>outputs</i>	<i>Processed data to storage</i>



3.1.5 DB store system.

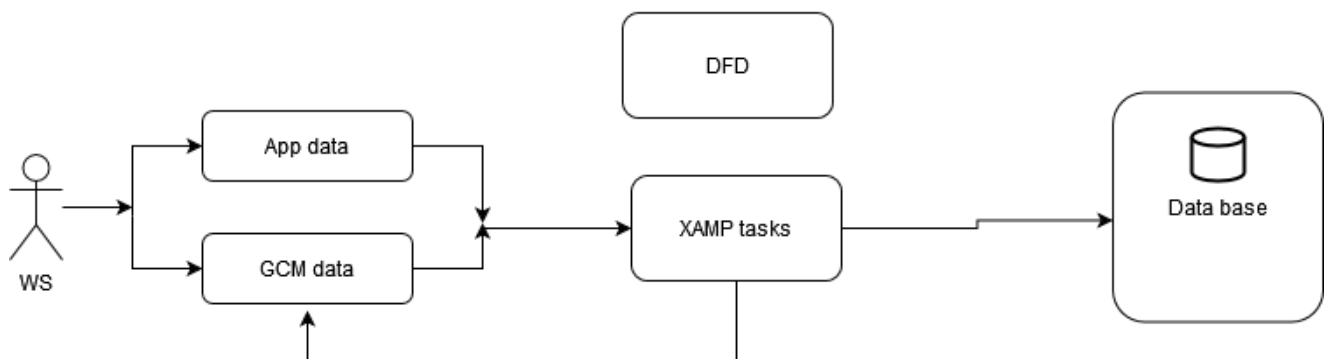
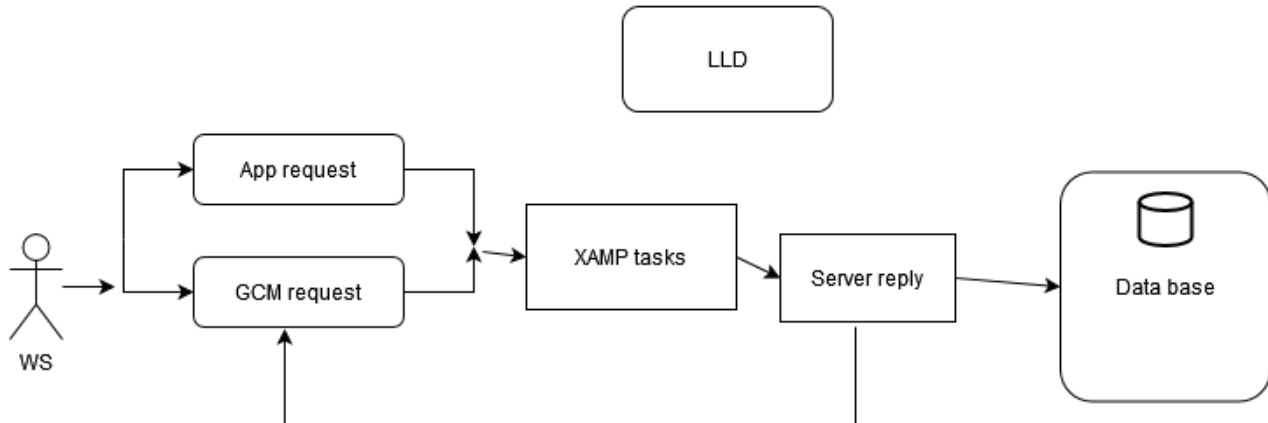
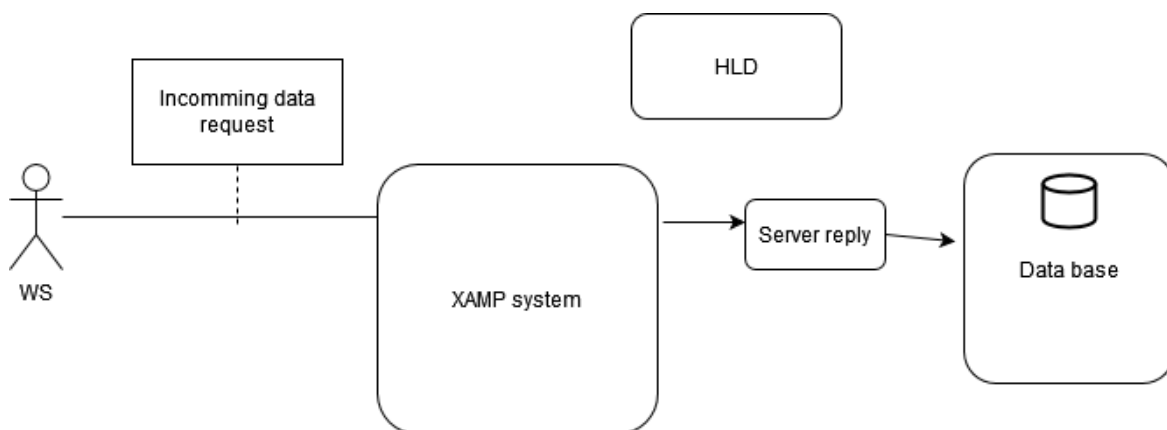
DB	
DB will receive data from all modules and store it, also give access to it to any module.	
<i>inputs</i>	<i>Data from adjacent modules</i>
<i>outputs</i>	<i>Warnings to adjacent modules</i> <i>Reports for user</i> <i>stored data for GCM</i>



3.1.6 Web server system.

This module will maintain the communication between the automated system and the user app.

Web server	
This module will maintain the communication between the automated system and the user app.	
<i>inputs</i>	<i>Requests from app</i>
<i>outputs</i>	<i>Replays from server, GCM and DB</i>



3.2 Use Cases:

3.2.1 Use Case “First Run”

ID	<i>CR-01</i>
Description	<i>Automation systems check</i>
Actors	<i>GCM, RCM and GHCM</i>
Preconditions	<i>GCM, RCM and GHCM software must have been installed in third party hardware</i>
Basic Steps	<i>1 Admin must check Connection between modules from GCM 2 Admin have to run “First Run” routine on GCM 3 Restart DB values to default</i>
Exception case	<i>If connection between modules fail Restores Software an retry</i>
Business validations/Rules	<i>Connection between modules must function GCM, RCM and GHCM software must have been installed Third party hardware must be functional</i>
Postconditions	<i>Rover must return to start point , Green house have to isolate both have to enter stand-by state</i>

4 Proposed documentation

Taking in mind the target of sift and the level of technology knowledge of them is proposed the develop of the next documentation:

- Quick start guide
- Full software install manual
- Support portal

5 Technical Requirements (Nonfunctional)

<i>Software used</i>	<i>C++ variant , linux</i>
<i>Operating system</i>	<i>linux</i>
<i>Database</i>	<i>My SQL linux variant</i>
<i>Front end Framework</i>	<i>Android studio</i>

5.2 Performance

GCM and sub modules must work 24/7

Task for modules have to be processed at least every 2 hours

User app have to be working 24/7

5.3 Scalability

after the first year of data retrieval sift can expand to 2 or 8 times its capacity it also can diverge to a research lab or to offer lab quality products , after 5 years or so and whit the process of the expanded data Sift is hope to evolve to a autonomous facility whit the help of a neural network or simplistic IA.

5.4 Security

GCM will have a security automatic routine "Isolation" to prevent damage to products after and attack to the network protocols or communication modules.

5.5 Maintainability

Software only need to be revised in the first run or at the end of a farm cycle.

5.6 Usability

Application can be used in androids devises whit android 6 or up.

5.7 Multi lingual Support

There should be option to convert change application language (Spanish, English, German to start).

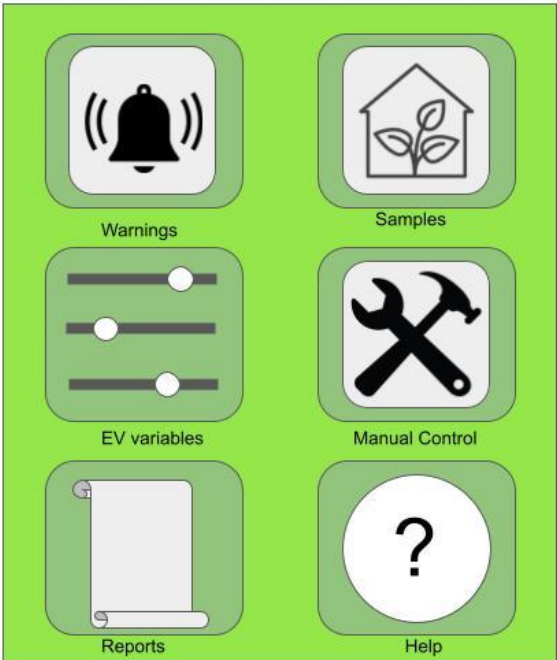
6 User app Prototype interface.



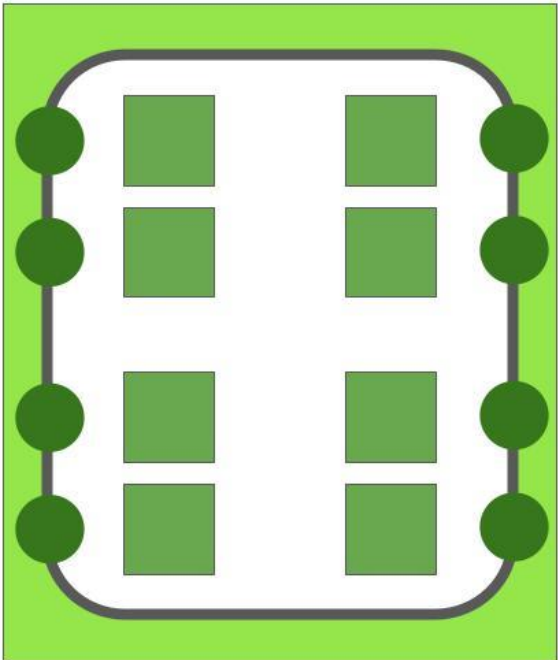
Start



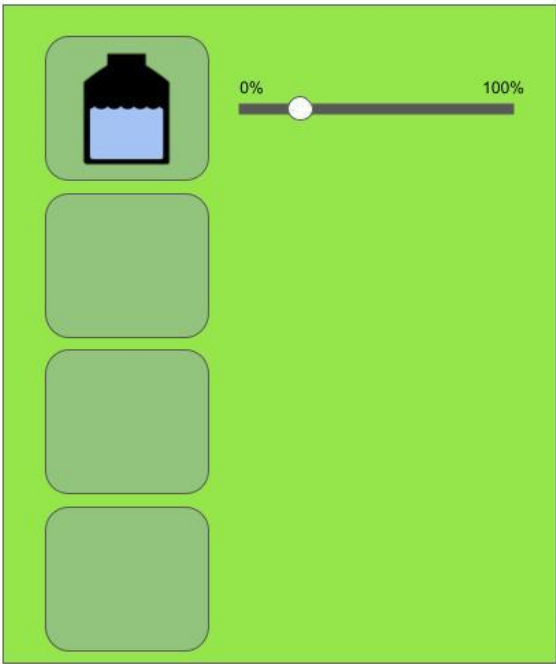
login



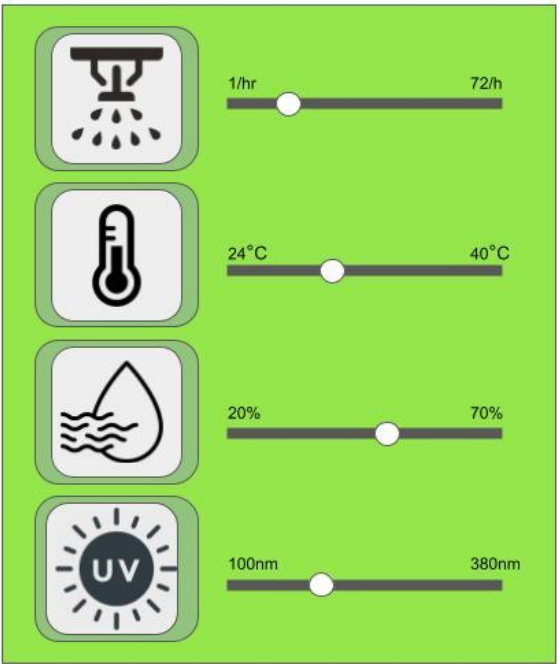
Utilities



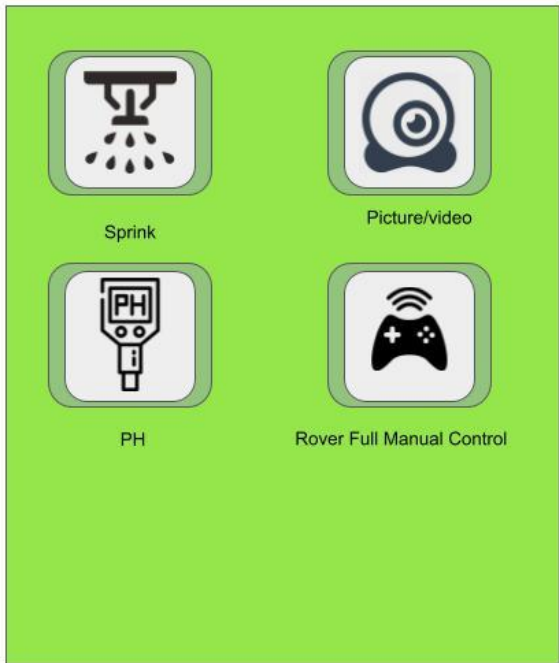
Samples



Warnings



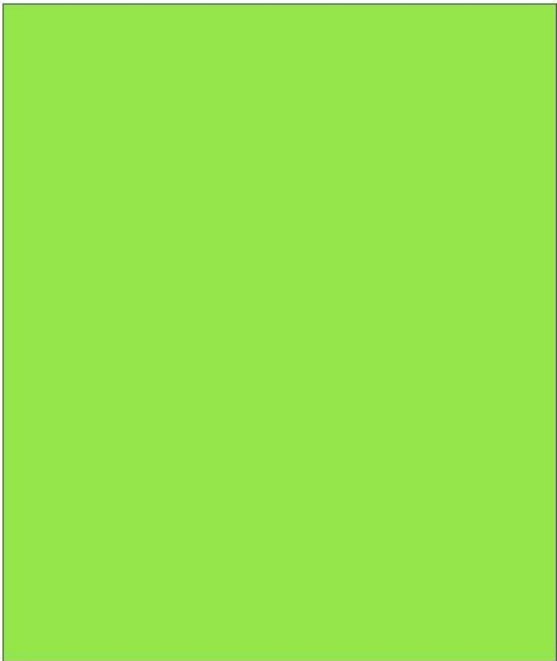
Environmental variables



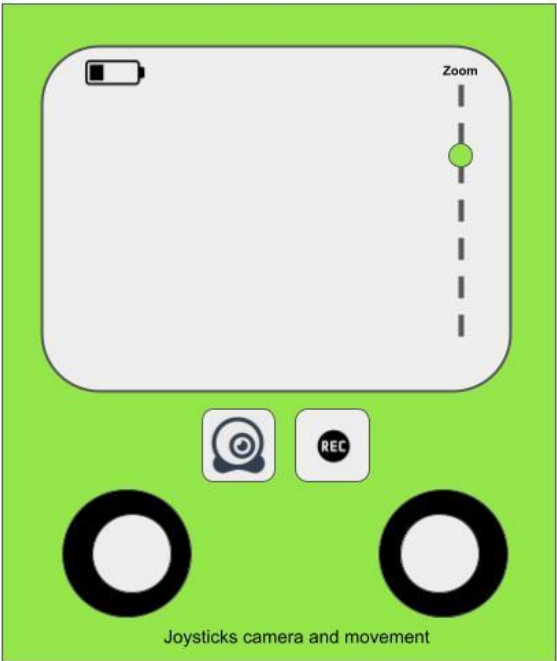
Manual Control



Reports



Help



Joysticks camera and movement

Full Manual Rover

7 Annexed document Full SOW document

STATEMENT OF WORK (Sif)

Autonomous farming system

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Issued By
Specialized Automated Solutions
Av. La Corona No. 320, Col La Palma, Loma la Palma, 00000 Ciudad de México
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INTRODUCTION

We aim to find a path to optimize and automate the process in which some edibles are cultivated and harvested.

BACKGROUND INFORMATION

in some regions of Europe exist a variety of berries that yet can't be cultivated by human labor, these can only be found in the wild. This project aims to study and find a way to change that situation by secluding the most the human factor and adding various ways to control a specific environment. By accomplish this the production of these products can be raised without depending of natural conditions.

CURRENT ENVIRONMENT

Current state.

- Develop a management semi-autonomous system for cultivate.
- App to manage actions and see registers.
- DB of instructions, sensors data and notifications.
- Semi-autonomous rover system and environmental control.
- User counts with a basic rover, a greenhouse and a basic irrigation system
- Budget is medium because the client wants a fast prototype instead a full develop system but some partners don't agree.
- The client and at last 3 more partner are involved.

GOALS AND OBJECTIVES

Business and Solution Objectives- ?

Technical Objectives- develop 2 semi-autonomous systems, integer them in a user interface, store environmental data.

Service Objectives- wireless access to data and system.

Security Objectives- Store/backup data and limited the access to it.

SCOPE OF WORK

From a high-level perspective, describe the project work and what it entails. Describe what is included. If helpful, also describe what is not included in the project work? Explain what will be accomplished. Describe the size of the effort. Are there any special areas of interest? Describe the methods of deliver.

DELIVERABLES

1. User system interface
2. DB for sensors data, instructions and users
3. Rover Semi-autonomous system
4. Greenhouse Semi-autonomous system
5. Final full integer system

MILESTONES

Milestone	Estimated Delivery Date
User access interface	18/04/2022
User controller's and data access modules	20/06/2022
DB user app modules	30/06/2022
Rover system	22/09/2022
DB rover data reception and instructions transmit.	18/10/2022
Greenhouse system	02/12/2022
DB Greenhouse data reception and instructions transmit.	25/12/2022
DB Store, access and backup system	10/01/2023
Full modules integration.	20/02/2023
Completion	15/03/2023

PERIOD OF PERFORMANCE

The project will have a duration around 6 months to 1 year after the client confirmation and at last a third of advance deposit. The project will be finished after a performance of the finished software and the final deposit by the client.

PLACE OF PERFORMANCE

For the project we will take lab 2 and 3 of the institution in Mexico City.

APPLICABLE STANDARDS

none yet

SPECIFIC REQUIREMENTS

check first part of integer document

RESOURCE REQUIREMENTS

check first part of integer document

HUMAN RESOURCES

Project Title	Required Knowledge/Skills
---------------	---------------------------

Data Server	DB developer
User interface	Graphic designer / app developer
Rover system /greenhose system	Microcontroller programming developer

OTHER RESOURCES

Resource Description
Private data server
Pre-built greenhouse or remote access to it

VENDOR RESPONSIBILITIES

- Release the documentation for the client
- Show advancement periodically
- Run test in the final deploy
- Don't surpass the budget

CLIENT RESPONSIBILITIES

- Do the payments
- Don't change drastically date of release
- Don't make big of absurd changes in the previous requirements

PROJECT RISKS

- The hardware already owned by the client may be malfunction
- Because of the distance and the third-party implementation there is a probability of malfunctions
- Because of the type of project any external (incidents) variables can affect the hardware.
- Because of the limited knowledge of the products or process a chance in a system may be needed

ASSUMPTIONS

- Client already own most of the hardware

- Client has a third party to deploy system
- Hardware functions fine
- Client is ready to deploy software in his/her hardware

COMPLETION CRITERIA

- At least 5 full positive tests of the final integer system
- Modules satisfy client requirements

CHANGE CONTROL PROCEDURE

- Appoint a meeting with the client
- Establish who wants the change
- If off limits, contract ends
- If not rearrange dates, budget and requirements

CONTRACT TYPE AND INVOICE PROCEDURES

Time and materials contract

OTHER INFORMATION AND SUPPORTING DOCUMENTATION

not yet established.

POINTS OF CONTACT

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Martinez Bautista Bryan Fernando bryan.martinez@estudiante.uacm.edu.mx phone: 5580412654 +52

ACCEPTANCE

Date: __yet to be accepted__

By initialing each page and signing below, I Heikkinen Eevi, in my capacity as General manager, of Sunrise berries farm agree to and accept the terms set forth in this Statement of Work.

Specialized Automated Solutions

By: _____ yet to be accepted _____
Signature

__ Heikkinen Eevi __
Printed Name and Title

8 Annexed document ROC prototypes



SIF

Organisation Specialized Automated Solutions
Department Business services

install request

SIF

Date: 06/05/2022

Doc. Version: [Status]

Change Request Form

SIF

Project:	<i>SIF</i>	Change ID:	<i>CG-01</i>
-----------------	------------	-------------------	--------------

Change Request	
Change Title:	<i>First install</i>
Requested by:	<i>Midnight Tea</i>
Identification Date:	<i>06/05/2022</i>
Category:	Services
Priority:	☐ Very High ☐ High ☒ Medium ☐ Low ☐ Very Low
Change Description & Details	
Request of miscellaneous service “first install” “first install” service will install of software components in the third-party hardware in the next order: - Rover module software - greenhouse module software - S.O. for server - General control module software - DB software and standard contents - Web server software - User app After install “first calls” routine will be run to check communication between modules and behaviour. lastly the requester will be notified depending on the result. Current Situation: Requester states inability to understand the process of install server environment even whit user manual also a lack of time. Requester only install third party hardware and primary services. Desired Situation: After fulfil the request the server environment along whit rover and greenhouse software will be ready and Business services request “first run” routine to check software behaviour in third party software. Impact or Risks: Report needs to state requester behaviour for future request also depending of the state of the hardware there will be an extra fee added to the cost of the “first install” service , “first run” routine is supposed to being execute by end user in this case an extra fee is added but as request of Business services this time it wont Out of Scope: Extra services like “manual test” or “peer” are out of the responsibilities, “first install” service won’t take responsibility for hardware error, primary services errors or extra request made the day of the change. Personal assistance is limited to give information of the change and will appoint to read the manual or make a future request to of Business services, support services or appoint a meeting.	

References and Related Documents	Location
SOW.doc	U:\SIF\Documents\



Organisation Specialized Automated Solutions
Department Support services

rollback request

SIF

Date: 06/05/2022
Doc. Version: [Status]

Change Request Form

SIF

Project:	SIF	Change ID:	CG-02
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Change Request	
Change Title:	rollback
Requested by:	Gaia Health
Identification Date:	06/05/2022
Category:	Services
Priority:	☐ Very High ☒ High ☐ Medium ☐ Low ☐ Very Low
Change Description & Details	
Request of support service "restore" "restore" service will do a rollback of the software components in conflict or bugged: - Rover module software will reload .bin archive to previous version - greenhouse module software will reload .bin archive to previous version - S.O. for server will reload or restore previous version - General control module software will load previous version - DB software and standard contents - Check "backup" routine execution before last update or change. - Take Bp.rar file from /home/user/Sif/DB/Backup/ - Restart DB version to previous - run "restore SGCD" routine appointing Bp.rar file - If still DB system working normal but behaviour is not correct run "soft restore" routine appointing Bp.rar file - Web server software restores previous settings - User app restore previous version - Process executed depend on the software behaviours After process "restore" routine run check behaviour between modules and communication. lastly the requester will be notified depending on the result. Current Situation: Requester states that rover module is behaving wrong it implies the module make abnormal task shutdown and not responding to manual control, it starts whit last system update, greenhouse module present some errors too, user app and web server are mainly behaving ok. Looking in the requester report it appears that the DB update has bugs and triggering the wrong behaviours in the rover and green house modules sending wrong data to the general control module Desired Situation: Support service "restore" for DB will be executed and check that the previous data are restore then "first run" routine be executed to check the rover and green house modules behaviour are correct. Impact or Risks: Hardware damage risk is high because rover module is autonomous and has high mobility it implies possible cost for hardware repair, requester products has high of damage depending in time response (possible trigger of "isolated" routine help) Out of Scope: Extra services like "manual test" or "peer" are out of the responsibilities, "restore" service won't take responsibility for hardware error, primary services errors or extra request made the day of the change. Personal assistance is limited to give information of the change and will appoint to read the manual or make a future request to of Business services, support services or appoint a meeting.	

References and Related Documents	Location
SOW.doc	U:\SIF\Documents\

9 Annexed document Documentation proposal (Spanish)

El proyecto denominado SIF que se esta desarrollando refiere a poder auxiliar en la rama de la industria referida como agronomía esta enfrenta 3 mayores problemas en este lapso de tiempo:

- Cambio climático (sería más bien colapso climático).
- El abandono de las fincas por las nuevas generaciones.
- Reducción de agua y suelos útiles

Tomando estos 3 problemas en consideración el Target de SIF se encontraría en personas con conocimientos de agronomía de los básico a casi avanzado pero teniendo en cuenta que estos podrían tener un acceso restringido a tecnología de alta gama dados los costos, pueden verse atraídos por la tecnología desarrollada hoy en día dado su desinterés por la industria a tratar , por lo cual trataremos su conocimiento en tecnología de básico a medio, tomaremos un primer rango de edad de 16 a 35 años pensando en que el usuario que administrara el sistema se encuentra en este y un posible rango superior de poder se encuentre en un rango mayor de edad. Teniendo como ultima consideración que el sistema pueda llegar a utilizarse en módulos pequeños de manera urbana y usuarios de esta demografía.

Teniendo en cuenta esta se propone que la documentación a desarrollar se base en el modelo ágil dado que el target del proyecto no requerirá información técnica específica y será capaz de administrar el sistema con ejemplos de funcionamiento y guías de usuario rápidas considerando que la interfaz de usuario es lo suficiente amigable. Así bien podemos inclinarnos a solo tener documentación de usuario dado que se espera que el usuario final instale el software de los módulos por sí mismo será necesario generar los 2 tipos de documentación de esta rama referidos como documentación de usuario final y documentación de administrador del sistema. Por lo tanto, se considera necesario el desarrollo de estos documentos:

- Manual completo de instalación de software
- Guía de inicio rápido
- Un portal de asistencia para futuras actualizaciones o resolución de problemas

Esta es el tipo de documentación que se cree necesaria para el proyecto, si algún usuario se ve interesado en escalar el proyecto al punto de poder ofrecer productos con calidad de laboratorio, hacer uso del mismo para investigación biotecnológica o similar se requerirá una reestructuración de la documentación, así como un cambio en los módulos referentes dependiendo de las necesidades del interesado.

10 Reserved for future changes

